How to use the MyForterro API

MyForterro distributes functionnalities to the Forterro products by providing a set of rest API endpoints. As of the time of this writing, the available modules are:

- **Fx Rates** - give real time exchange rates for a base currency
- **E-Invoicing** - set of API to deal with the national laws regarding digital invoices
- **AI** - set of APIs to access the Forterro inference API and the agentic platform
- (Open Banking) - In construction

You can browse the integration environment API using this [swagger UI](#).

Calling the API

To call the API, two things are needed:

- a **MyForterro JWT access token**, obtained using an oauth 2.X flow. This token is passed in the `authorization` http header as a `Bearer` token.
- a **tenant ID** - every call to the API is done on a specific tenant. Usually, a tenant corresponds to one customer. The tenant id is passed using the `MFT-Tenant-Id` http header.

Authentication and authorization

The MyForterro API services are secured by the MyForterro core authentication and authorization services. Each tenant in the MyForterro API has its counterpart in the core platform in the form of a ***MyForterro business service application***. This application holds the secret wich will be needed by the tenant to secure the access. When an admin configures the service, the is and secrets are configured on the tenant.

A user or an application, in order to connect to the tenant, will need to have some specific **roles** on this application. The roles needed will depends on the modules which the endpoints belong to:

- Fx Rates - no role needed
- E-Invoicing - Accounting or admin role
- AI - User or Admin role
- Open Banking - WIP

In addition, the **module** in question will need to be enabled in the application overview page.

Note: the MyForterro API services checks the modules configuration every 5' for performance reasons

Getting a token

An access token will need to be created for the specific target tenant in order to access the API. This is done by providing the `application` parameter in the oauth flow, containing the application ID of the corresponding application. By doing so, MyForterro is requested to provide a token targetting the appropriate audience, and to include all existing roles and licenses for the user or application.

Example: getting a token in client credentials

💡 This python code example is targetting the integration environment.

```python
import requests

TOKEN_URL = "https://integration-myforterro-core.fcs-dev.eks.forterro.com/connect/token"
CLIENT_ID = "the MyForterro client application ID"
CLIENT_SECRET = "the MyForterro application client Secret"
TARGET_APPLICATION = "the MyForterro tenant application ID"

data = {
    "grant_type": "client_credentials",
    "client_id": CLIENT_ID,
    "client_secret": CLIENT_SECRET,
    "application": TARGET_APPLICATION,
}

response = requests.post(TOKEN_URL, data=data)

if response.status_code == 200:
    token = response.json().get("access_token")
    print("Access Token:", token)
else:
    print("Error:", response.status_code, response.text)

```

Example: calling the inference API with an OpenAI sdk

💡 This python code example is targetting the integration environment.

```python
#!/Library/Frameworks/Python.framework/Versions/3.12/bin/python3

import openai
import requests

API_BASE = "https://integration-myforterro-api.fcs-dev.eks.forterro.com/v1/ai/inference/openai"
API_KEY = "your MyForterro access token"
TENANT_ID = "your tenant ID"

# Configure OpenAI client
client = openai.OpenAI(
    api_key=API_KEY, base_url=API_BASE, default_headers={"MFT-Tenant-Id": TENANT_ID}
)

response = client.chat.completions.create(
    model="gpt-4o",
    messages=[
        {
            "content": "You are an helpful assistant.",
            "role": "system",
        },
        {
          "role": "user",
          "content": "Tell me a joke."},
    ],
    temperature=1.0
)

print(f"🤖> {response.choices[0].message.content}")

```